

Ashwin Prabou

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EDUCATION

University of California, Santa Cruz

June 2027

B.S. Computer Science

GPA: 3.6

Coursework: Data Structures & Algorithms, Computer Systems, Databases, Embedded Operating Systems, Agile Development, Machine Learning, Probability & Statistics, Compiler Design

EXPERIENCE

CruzHacks

June 2025 – Present

Engineering Director

Santa Cruz, CA

- Led cross-functional collaboration with design and operations stakeholders to deliver a **scalable full-stack platform** serving **500+** users, supporting registration, team formation, and live judging workflows.
- Architected **backend services and REST APIs** using TypeScript, Node.js, and Firebase, improving performance by **35%** and ensuring data consistency under peak event load.
- Designed CI/CD pipelines with automated testing, monitoring hooks, and structured logging, increasing deployment reliability and reducing production bugs by **70%** during live events.
- Improved system **observability and availability** by integrating performance tracking and validation checks across build and release workflows.

Project Terraforma

Sep. 2025 – Dec. 2025

Open Source Contributor

Santa Cruz, CA

- Engineered a **Python-based ETL pipeline** processing **2.3GB** of geospatial datasets using Pandas, enabling structured compression and reliable downstream analytics workflows.
- Built a full-stack analytics platform with a **Flask REST API** and SQLite backend, delivering real-time dashboards and conditional feedback systems for non-technical users.
- Designed automated validation checks across **9** multi-release datasets, generating schema metadata and statistical summaries to improve dataset transparency, reliability, and reproducibility.

Perfint Healthcare

Jan. 2025 – June 2025

Software Engineering Intern

Remote

- Developed a C++ RRT* path planning algorithm, improving routing accuracy by **15%** for CT-guided medical interventions in a regulated clinical environment.
- Collaborated in Agile sprints with cross-functional teams to refine requirements, conduct structured code reviews, and validate algorithmic performance and safety constraints.
- Contributed to validation workflows emphasizing **correctness, reproducibility, and reliability** in safety-critical medical software systems.

PROJECTS

LeetStreak | AI-Powered Skill Coaching Platform

May 2025 – Present

- Developed a **full-stack, containerized web application** enabling collaborative LeetCode progress tracking with authentication, analytics, and social features.
- Designed and integrated an **AI-powered Skill Coach** that analyzes user telemetry (problem difficulty, tags, solve rate, time trends) to generate personalized learning insights and adaptive practice recommendations.
- Engineered feature extraction pipelines to compute topic-level proficiency metrics and stagnation signals, feeding structured data into an LLM for contextual feedback generation.
- Built structured **evaluation tooling** to assess AI output quality using heuristic scoring, response constraints, and consistency checks, incorporating user feedback signals to iteratively refine prompt design.
- Implemented telemetry logging for prompt versions, response metrics, and user engagement analytics, enabling data-driven optimization of AI-generated learning experiences.
- Applied **Responsible AI principles** by enforcing output templates, limiting speculative claims, and introducing deterministic fallback recommendations to reduce hallucination risk.

SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL

AI & Machine Learning: LLM integration, Prompt Engineering, Model Evaluation (Precision/Recall/F1), Feature Engineering, scikit-learn, AI-assisted Development (GitHub Copilot)

Backend & Services: RESTful APIs, Flask, Node.js, Firebase, Authentication, Data Modeling

Frontend: React, Next.js, HTML5, CSS3

Cloud & DevOps: Firebase, Docker, CI/CD (GitHub Actions), Monitoring, Structured Logging

Data & Analytics: PostgreSQL, SQLite, Pandas, NumPy, ETL Pipelines

Testing & Tooling: Automated Testing, API Validation, Git/GitHub, Shell Scripting